

STOCK MARKET COMPLETE GUIDE

From Basics to Advanced Market Intelligence

15 Comprehensive Chapters
Covering Fundamentals · Analytics · Strategy · Psychology

Table of Contents

Chapter 01	—	Stock Market Basics
Chapter 02	—	Financial Metrics
Chapter 03	—	Country Regulations
Chapter 04	—	Exchanges & Trading
Chapter 05	—	Historical Market Events
Chapter 06	—	Sector Analysis
Chapter 07	—	SQL Analytics
Chapter 08	—	Investment Concepts
Chapter 09	—	Risk Management
Chapter 10	—	AI Market Trends
Chapter 11	—	Technical Indicators
Chapter 12	—	Macroeconomic Impacts
Chapter 13	—	Portfolio Strategies
Chapter 14	—	Market Psychology
Chapter 15	—	Real-World Case Studies

Chapter 1 — Stock Market Basics

The stock market is a regulated marketplace where buyers and sellers trade shares of publicly listed companies. It serves as a barometer of economic health and a vehicle for wealth creation over time.

What Is a Stock?

A stock (or share) represents a unit of ownership in a corporation. Stockholders are entitled to a proportional claim on the company's assets and earnings. Stocks are broadly classified into:

- Common Stock — carries voting rights and variable dividends
- Preferred Stock — fixed dividends, priority in liquidation, limited voting rights
- Growth Stocks — high reinvestment, low/no dividends, capital appreciation focus
- Dividend Stocks — regular income distributions to shareholders

How Stock Markets Work

Markets operate through a continuous auction mechanism. Buyers post bids; sellers post asks. A trade occurs when $\text{bid} \geq \text{ask}$. Electronic order-matching engines at exchanges execute millions of such trades per second.

Primary vs Secondary Market

Market	Purpose	Example
Primary	Companies raise capital via IPO	NSE IPO listing
Secondary	Investors trade existing shares	BSE daily trading
OTC	Unlisted securities traded directly	Penny stocks, bonds

Key Market Participants

- Retail Investors — individual traders using personal capital
- Institutional Investors — mutual funds, pension funds, insurance companies
- Market Makers — provide liquidity by quoting buy/sell prices
- Brokers & Dealers — intermediaries facilitating trades
- Regulators — SEBI (India), SEC (USA), FCA (UK)

Chapter 2 — Financial Metrics

Financial metrics allow investors to evaluate company performance, compare peers, and assess valuation. Mastering these ratios is foundational to fundamental analysis.

Valuation Ratios

Metric	Formula	Interpretation
P/E Ratio	Price / EPS	Higher = growth expectations or overvaluation
P/B Ratio	Price / Book Value	< 1 may indicate undervaluation
EV/EBITDA	Enterprise Value / EBITDA	Popular for M&A comparisons
PEG Ratio	P/E / Earnings Growth Rate	< 1 generally considered attractive
Dividend Yield	Annual Dividend / Price	Income potential measure

Profitability Metrics

- Return on Equity (ROE) = Net Income / Shareholders' Equity — measures management efficiency
- Return on Assets (ROA) = Net Income / Total Assets — asset utilization
- Net Profit Margin = Net Income / Revenue — bottom-line efficiency
- EBITDA Margin = EBITDA / Revenue — operational profitability
- Earnings Per Share (EPS) = Net Income / Shares Outstanding

Liquidity & Leverage

- Current Ratio = Current Assets / Current Liabilities (>1 healthy)
- Quick Ratio = (Current Assets - Inventory) / Current Liabilities
- Debt-to-Equity = Total Debt / Equity (sector-specific benchmarks apply)
- Interest Coverage = EBIT / Interest Expense (>3x preferred)

Chapter 3 — Country Regulations

Every country's stock market operates under its own legal and regulatory framework designed to protect investors, ensure fair trading, and maintain market integrity.

India — SEBI

Securities and Exchange Board of India regulates all exchanges. Key rules: Insider Trading Regulations 2015, LODR (Listing Obligations & Disclosure Requirements), T+1 settlement cycle.

USA — SEC

Securities and Exchange Commission enforces Dodd-Frank Act, Sarbanes-Oxley Act. FINRA oversees broker-dealers. Regulation NMS governs equity trading.

UK — FCA

Financial Conduct Authority under FCA Handbook. Post-Brexit adaptation of MiFID II rules. Senior Managers Certification Regime (SMCR).

EU — ESMA

European Securities and Markets Authority coordinates MiFID II / MiFIR across 27 nations. GDPR intersects with financial data reporting.

China — CSRC

China Securities Regulatory Commission. A-shares (domestic), B-shares, H-shares (HK listed). Stock Connect programs link mainland with HK.

Japan — FSA

Financial Services Agency. Tokyo Stock Exchange rules. Stewardship Code promotes institutional investor engagement.

Universal Regulatory Themes

- Disclosure requirements — mandatory filing of financials
- Anti-money laundering (AML) and KYC norms
- Circuit breakers to halt trading during extreme volatility
- Short-selling restrictions in crisis periods
- Cross-border cooperation via IOSCO

Chapter 4 — Exchanges & Trading

Stock exchanges are centralized platforms (physical or electronic) that facilitate transparent, regulated securities trading. Modern exchanges are almost entirely electronic.

Major Global Exchanges

Exchange	Country	Index	Market Cap (approx)
NYSE	USA	Dow Jones, S&P 500	~\$25T
NASDAQ	USA	NASDAQ Composite	~\$22T
BSE	India	SENSEX	~\$4T
NSE	India	NIFTY 50	~\$4T
Shanghai SE	China	SSE Composite	~\$7T
LSE	UK	FTSE 100	~\$3.8T
Euronext	EU	CAC 40, AEX	~\$6T
TSE (JPX)	Japan	Nikkei 225	~\$5.5T

Order Types

- Market Order — execute immediately at best available price
- Limit Order — execute only at specified price or better
- Stop-Loss Order — triggers market order when price hits stop level
- Stop-Limit Order — triggers limit order at stop level
- Good Till Cancelled (GTC) — remains active until manually cancelled
- Iceberg Order — large order split into visible + hidden portions

Trading Sessions

- Pre-market session — price discovery before official open
- Regular session — main trading hours (e.g., NSE: 9:15 AM – 3:30 PM IST)
- After-hours session — extended trading post-close
- T+1 / T+2 Settlement — trades settle 1–2 days after execution

Chapter 5 — Historical Market Events

Understanding historical market crises helps investors recognize warning signs, manage risk, and appreciate market resilience over long periods.

1929 — The Great Crash

Wall Street Crash triggered by excessive speculation and leverage. Dow fell ~89% from peak to trough. Led to the Great Depression and establishment of the SEC in 1934.

1987 — Black Monday

October 19: Dow fell 22.6% in a single day — the largest one-day percentage drop ever. Caused by portfolio insurance programs and automated selling cascades. Led to introduction of circuit breakers.

2000 — Dot-com Bubble

NASDAQ rose 400% from 1995–2000 driven by internet euphoria. Collapsed 78% by 2002. ~\$5 trillion in market cap wiped out. Many unprofitable startups went bankrupt.

2008 — Global Financial Crisis

Subprime mortgage collapse triggered systemic banking failure. Lehman Brothers bankruptcy. S&P; 500 fell 57%. TARP (\$700B) bailout and global QE followed.

2010 — Flash Crash

May 6: Dow plunged ~1,000 points in minutes then recovered. Caused by high-frequency trading feedback loops. Led to new market circuit breaker rules.

2020 — COVID-19 Crash

S&P; 500 fell 34% in 33 days — fastest bear market ever. Massive stimulus (\$5T+) drove one of the fastest recoveries. Remote-work tech stocks soared.

Key Lessons

- Markets always recover given sufficient time
- Leverage amplifies both gains and losses catastrophically
- Regulatory responses shape future market structure
- Sentiment cycles between greed and fear predictably
- Diversification reduces but does not eliminate systemic risk

Chapter 6 — Sector Analysis

GICS (Global Industry Classification Standard) divides the market into 11 sectors. Sector rotation — the movement of capital between sectors — is a core strategy in portfolio management.

Information Technology

Semiconductors, software, IT services. High growth, sensitive to interest rates. Key metric: P/E, revenue growth.

Healthcare

Pharma, biotech, devices, services. Defensive nature, FDA approvals drive volatility.

Financials

Banks, insurance, asset managers. Sensitive to interest rate changes and credit cycles.

Consumer Discretionary

Retail, autos, media. Cyclical — performs well in economic expansions.

Consumer Staples

Food, beverages, household products. Defensive — resilient in downturns.

Energy

Oil & gas, renewables. Tied to commodity prices and geopolitical events.

Industrials

Manufacturing, aerospace, logistics. Sensitive to GDP growth.

Materials

Mining, chemicals, construction materials. Commodity-price driven.

Utilities

Electric, gas, water. Rate-sensitive, high dividend yield, defensive.

Real Estate

REITs, property developers. Sensitive to interest rates and credit availability.

Communication Services

Telecom, media, internet platforms. Mix of growth and value characteristics.

Chapter 7 — SQL Analytics

SQL (Structured Query Language) is widely used in finance for querying market databases, calculating metrics, and building trading dashboards.

Common Financial SQL Queries

Top Performing Stocks by Return

```
SELECT ticker, ((close_price - open_price) / open_price) * 100 AS pct_return
FROM daily_prices
WHERE trade_date = '2024-01-15'
ORDER BY pct_return DESC LIMIT 10;
```

52-Week High/Low

```
SELECT ticker,
MAX(high_price) AS week52_high,
MIN(low_price) AS week52_low
FROM daily_prices
WHERE trade_date >= CURRENT_DATE - INTERVAL '52 weeks'
GROUP BY ticker;
```

Moving Average (20-day)

```
SELECT ticker, trade_date,
AVG(close_price) OVER (
PARTITION BY ticker
ORDER BY trade_date
ROWS BETWEEN 19 PRECEDING AND CURRENT ROW
) AS ma_20
FROM daily_prices;
```

Sector Aggregation

```
SELECT sector, COUNT(*) AS stocks,
AVG(pe_ratio) AS avg_pe,
SUM(market_cap) AS total_market_cap
FROM company_fundamentals
GROUP BY sector
ORDER BY total_market_cap DESC;
```

Useful SQL Concepts for Market Data

- Window functions (LAG, LEAD, RANK) for time-series analysis
- CTEs for readable multi-step queries
- Partitioning for per-ticker calculations
- Date functions for rolling windows and period comparisons
- JSON columns for storing OHLCV tick data efficiently

Chapter 8 — Investment Concepts

Sound investment requires understanding core frameworks that guide decision-making across asset classes, time horizons, and market conditions.

Value Investing

Pioneered by Graham & Buffett. Buy stocks trading below intrinsic value (margin of safety). Focus on P/B, P/E, free cash flow. Long-term buy-and-hold orientation.

Growth Investing

Target companies with above-average revenue/earnings growth. Typically higher P/E tolerated. Examples: early Amazon, Google, Infosys.

Dividend Investing

Build portfolio of high-yield, consistent dividend payers. Reinvest dividends (DRIP) for compounding. Focus on dividend coverage ratio and payout sustainability.

Index Investing

Passive strategy tracking benchmark indices. Low cost, diversified, market-return objective. Championed by John Bogle (Vanguard).

Momentum Investing

Buy recent winners, sell recent losers. Based on persistence of price trends. Requires disciplined entry/exit rules.

Contrarian Investing

Bet against prevailing market sentiment. Buy fear, sell greed. Requires patience and high conviction.

Dollar-Cost Averaging (DCA)

Invest fixed amount at regular intervals regardless of price. Reduces impact of volatility and removes market-timing emotion.

Compounding

Einstein's 'eighth wonder' — returns generate returns. Rs 1 lakh at 12% CAGR = Rs 17.4 lakh in 25 years.

Chapter 9 — Risk Management

Risk management is the systematic process of identifying, assessing, and mitigating financial risks. It is what separates professional investors from gamblers.

Types of Risk

- Market Risk — prices move against your position (systematic risk)
- Credit Risk — counterparty defaults on obligation
- Liquidity Risk — unable to exit position at fair price
- Concentration Risk — over-exposure to single stock/sector
- Operational Risk — systems, processes, human error
- Regulatory/Legal Risk — adverse policy or litigation
- Currency Risk — forex movements impact returns

Key Risk Metrics

Metric	Description	Target
Beta	Sensitivity to market moves	< 1 = defensive
Sharpe Ratio	Return per unit of risk	> 1 good, > 2 excellent
Max Drawdown	Peak-to-trough portfolio decline	Minimize
VaR (95%)	Max loss in 95% of scenarios	Quantify & cap
Standard Deviation	Volatility of returns	Lower = stable
Sortino Ratio	Return / downside deviation	Higher = better

Risk Control Strategies

- Position sizing — never risk more than 1–2% of capital on single trade
- Stop-loss orders — automate exit at predefined loss level
- Portfolio diversification across sectors, geographies, asset classes
- Hedging with options (puts) or inverse ETFs
- Regular portfolio rebalancing to maintain target allocation
- Scenario analysis and stress testing

Chapter 10 — AI Market Trends

Artificial Intelligence is fundamentally transforming financial markets — from trading algorithms to compliance automation and investor sentiment analysis.

Algorithmic & HFT Trading

AI-driven algorithms execute trades in microseconds. High-Frequency Trading (HFT) accounts for 50–70% of US equity volume. Uses ML for pattern detection, arbitrage, and market making.

Natural Language Processing (NLP)

Sentiment analysis of news, earnings calls, social media (Twitter/Reddit) to predict price movements. LLMs summarize analyst reports and SEC filings instantly.

Predictive Analytics

ML models (LSTM, XGBoost, Transformer-based) trained on OHLCV data, macro indicators, and alternative data for return prediction.

Alternative Data

Satellite imagery (retail foot traffic), credit card transaction data, web scraping, app download trends. Alpha generation edge over traditional fundamental analysis.

Robo-Advisors

Automated portfolio management (Betterment, Wealthfront, Zerodha Kite). ML-driven rebalancing, tax-loss harvesting, risk profiling.

AI in Risk Management

Real-time fraud detection, credit risk scoring, stress testing simulations with neural networks.

RegTech

AI automates compliance reporting, AML transaction monitoring, and regulatory filing at scale.

LLM-Powered Research

GPT-class models synthesize vast research corpora, generate earnings previews, and assist portfolio managers in decision-making.

Chapter 11 — Technical Indicators

Technical analysis uses price, volume, and derived indicators to forecast future price movements. It assumes all information is already reflected in the price.

Trend Indicators

Moving Average (MA)

SMA: average over N periods. EMA: exponentially weighted. Crossovers (Golden Cross / Death Cross) signal trend changes.

MACD

Moving Average Convergence Divergence. MACD line = 12-EMA minus 26-EMA. Signal line = 9-EMA of MACD. Histogram shows momentum.

Parabolic SAR

Trailing stop-and-reverse indicator. Dots below price = uptrend; dots above = downtrend.

Bollinger Bands

20-day SMA $\pm$ 2 standard deviations. Price touching upper band = overbought signal; lower band = oversold.

Momentum / Oscillators

RSI (Relative Strength Index)

0–100 scale. Above 70 = overbought; below 30 = oversold. Divergence with price is powerful signal.

Stochastic Oscillator

%K and %D lines. Compares close to range over N periods. Works well in ranging markets.

CCI (Commodity Channel Index)

Measures deviation from statistical mean. ± 100 thresholds used for signals.

Volume-Weighted Average Price (VWAP)

Intraday benchmark. Institutional orders benchmarked to VWAP. Trading above = bullish.

Support & Resistance

- Horizontal levels — price zones where buying/selling repeatedly emerges
- Trendlines — connect swing highs (resistance) or lows (support)
- Fibonacci retracements — 23.6%, 38.2%, 50%, 61.8% levels
- Pivot Points — calculated from previous day's OHLC

Chapter 12 — Macroeconomic Impacts

Macroeconomic forces drive broad market trends. Understanding their transmission mechanisms is essential for top-down investment analysis.

Interest Rates

Central bank rate hikes increase discount rates, reducing equity valuations (DCF impact). Banks benefit; growth stocks suffer. Rate cuts stimulate markets. Fed Funds Rate, RBI Repo Rate are key benchmarks.

Inflation

Moderate inflation (2–3%) is healthy. High inflation erodes purchasing power, triggers rate hikes. Real assets (commodities, real estate, TIPS) hedge inflation. CPI, WPI, PCE are key measures.

GDP Growth

Positive GDP growth supports corporate earnings. Contraction (2+ quarters) = recession. Leading indicators: PMI, consumer confidence, building permits.

Currency Exchange Rates

Weak domestic currency boosts exporters (IT, pharma in India) but increases import costs. DXY (US Dollar Index) moves impact global risk appetite.

Unemployment Rate

Low unemployment signals economic strength but may trigger wage inflation. Non-Farm Payrolls (NFP) is the most market-moving US data release.

Fiscal Policy

Government spending and taxation. Stimulus packages boost demand; austerity suppresses it. Fiscal deficit levels affect bond yields and currency.

Global Trade & Geopolitics

Trade wars, tariffs, sanctions, and conflicts disrupt supply chains and commodity prices. Russia-Ukraine war reshaped energy markets; US-China tensions shaped semiconductor supply.

Key Economic Calendars Events

- US Federal Reserve FOMC meetings (8 per year)
- RBI Monetary Policy Committee (MPC) — bi-monthly
- US Non-Farm Payrolls — first Friday of each month
- CPI/WPI data releases — monthly
- Quarterly GDP estimates and revisions
- Corporate earnings seasons — quarterly

Chapter 13 — Portfolio Strategies

Portfolio strategy encompasses asset allocation, diversification, rebalancing, and the systematic translation of investment thesis into holdings.

Modern Portfolio Theory (MPT)

Harry Markowitz (1952). Efficient frontier — maximize return for given risk level. Correlation between assets drives diversification benefit. Mean-variance optimization.

Core-Satellite Approach

Core (60–80%): low-cost index funds. Satellite (20–40%): active picks, thematic ETFs, alternatives. Balances stability with alpha potential.

60/40 Portfolio

60% equities, 40% bonds. Traditional balanced allocation. Bonds cushion equity drawdowns. Lower effectiveness in recent rising-rate environments.

Factor Investing (Smart Beta)

Systematic exposure to proven return drivers: Value, Momentum, Quality, Low Volatility, Small Cap. Rules-based, between active and passive.

All-Weather Portfolio (Ray Dalio)

30% equities, 40% long-term bonds, 15% intermediate bonds, 7.5% gold, 7.5% commodities. Designed to perform across all economic environments.

Barbell Strategy (Nassim Taleb)

90% ultra-safe assets + 10% high-risk/high-reward bets. Avoid middle-risk assets. Protects against black swans while preserving upside.

Thematic Investing

AI, clean energy, digital payments, aging demographics. Concentrated bets on structural mega-trends. Higher conviction + higher risk.

Rebalancing

- Calendar rebalancing — quarterly or annually regardless of drift
- Threshold rebalancing — rebalance when allocation drifts $\pm 5\%$ from target
- Tax-efficient rebalancing — use new contributions to rebalance before selling
- Rebalancing captures value from mean reversion in asset classes

Chapter 14 — Market Psychology

Behavioral finance reveals that markets are driven not only by logic but by human emotions, cognitive biases, and collective psychology. Understanding these helps investors avoid costly mistakes.

Loss Aversion

Losses feel ~2x more painful than equivalent gains feel good (Kahneman & Tversky). Leads to holding losers too long and selling winners too early.

Confirmation Bias

Seeking information that confirms existing beliefs. Investors ignore contradictory data about their holdings.

Herd Behavior

Following the crowd. Creates bubbles (everyone buys) and crashes (everyone sells). FOMO (Fear of Missing Out) and panic selling are manifestations.

Anchoring

Over-reliance on first piece of information (e.g., original purchase price). Prevents rational reassessment of position.

Overconfidence Bias

Investors overestimate their analytical ability. Active traders underperform markets partly due to overconfidence-driven excess trading.

Recency Bias

Overweighting recent events. After a crash, investors are overly fearful; after a bull run, overly optimistic.

Disposition Effect

Tendency to sell winners too early (lock in gains) and hold losers too long (avoid realizing loss).

Mental Accounting

Treating money differently based on its source or intended use. Violates fungibility of money.

The Market Cycle of Emotions

- Optimism → Excitement → Thrill → Euphoria (market top)
- Anxiety → Denial → Fear → Desperation → Panic
- Capitulation → Despondency → Depression (market bottom)
- Hope → Relief → Optimism (recovery begins)

Strategies to Overcome Bias

- Pre-commit to investment rules via Investment Policy Statement
- Automate investments to remove emotional decision-making
- Keep an investment journal to track decision rationale
- Seek disconfirming evidence before making decisions
- Work with a financial advisor for accountability

Chapter 15 — Real-World Case Studies

Case studies distill complex market events into actionable lessons. Each case below highlights a different facet of market dynamics.

Case Study 1: Warren Buffett & Coca-Cola (1988)

Buffett purchased \$1B of Coca-Cola in 1988–1989 at ~15x earnings. Thesis: strong brand moat, global expansion potential, consistent free cash flow. By 2024, the position is worth ~\$25B with \$700M+ annual dividends. Lesson: identify durable competitive advantages and hold through volatility.

Case Study 2: Harshad Mehta Scam (India, 1992)

Broker Harshad Mehta exploited loopholes in bank receipt (BR) system to divert ~Rs 4,000 crore from banks to stock market. SENSEX rose 143% then crashed 72%. Led to SEBI strengthening, dematerialization of shares, and banking reforms. Lesson: regulatory gaps create systemic risk.

Case Study 3: Enron Collapse (2001)

Enron used special purpose entities (SPEs) to hide \$1B+ in debt off-balance-sheet. Stock fell from \$90 to \$0.67. 20,000 employees lost jobs and pensions. Led to Sarbanes-Oxley Act (SOX) requiring CEO/CFO to personally certify financials. Lesson: analyze financial footnotes; be skeptical of complex accounting.

Case Study 4: Tesla's Meteoric Rise (2019–2021)

Tesla rose 1,000%+ driven by EV adoption narrative, S&P 500 inclusion, retail investor enthusiasm, and short-squeeze dynamics. P/E reached 1,000x. Demonstrated power of narrative and momentum investing. Also showed risk: stock fell 65% in 2022. Lesson: sentiment can sustain irrational valuations temporarily.

Case Study 5: Infosys — India's IT Growth Story

Infosys went public at Rs 95 in 1993. By 2000 it reached Rs 9,000+ (split-adjusted). US listing on NASDAQ 1999. Revenue grew from \$2M (1991) to \$18B+ (2024). Lesson: quality management, scalable business model, and demographic tailwinds compound over decades.

Case Study 6: GameStop Short Squeeze (2021)

Reddit community r/WallStreetBets coordinated buying of heavily-shortaged GameStop (GME). Stock rose from \$20 to \$483 in weeks. Short-sellers (Melvin Capital) lost billions. Robinhood restricted buying amid clearing-house capital requirements. Lesson: social media can overwhelm institutional positions; liquidity constraints matter.

End of Guide

This guide covered 15 chapters spanning stock market fundamentals, financial metrics, global regulations, trading mechanics, historical events, sector dynamics, SQL analytics, investment strategies, risk management, AI trends, technical indicators, macroeconomics, portfolio construction, behavioral finance, and real-world case studies.

Invest wisely. Think long-term. Manage risk always.